PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Table of Contents

1. Project Overview 3
2. Background & Business Objective 4
3. Product Purpose 5
4. System Functionalities 6
5. User Stories & Use Cases 8
6. Features Included (Scope Definition) 10
7. Features Not Included (Scope Control) 10
8. Assumptions & Constraints 11
9. Risks & Questions Throughout Development 11

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Project Name: Content Buddy

Version: 1.0

1. Project Overview

- **Problem Statement:** High-quality social media content nowadays demands a good idea coupled with consistency of brand voice in spite of all the peculiarities of platform algorithms. Many creators and entrepreneurs find it hard to replicate the success they had in the past using their own viral style in new videos due to wasting too much time on writing scripts incompatible with their own personality. Moreover, keeping track of fast-changing trends and strict platform policies, such as shadowbanning certain words or copyright violations, is a huge obstacle.

- **Target Domain:** AI-based content strategizing and branding for TikTok, Instagram Reels, and YouTube Shorts.

- **Proposed Solution Summary:** Content Buddy is an AI-powered content strategist that transforms raw ideas into platform-compliant, trend-aligned content by analyzing a user's past work to replicate their unique tone, pacing, and style. It streamlines a traditionally time-consuming workflow into a single automated process, generating personalized scripts, storyboards, and posting schedules while recommending trending elements such as music and formats. In addition, it acts as a built-in compliance guard by continuously checking platform policies (e.g., TikTok, Instagram, YouTube) to prevent copyright and guideline violations. Overall, Content Buddy helps creators, students, and businesses save time, maintain brand consistency, and consistently produce relevant, high-quality content.

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2. Background & Business Objective

- **Background of the Problem:** Social media plays a powerful role in influencing behavior and has become essential for modern marketing, yet the content creation process remains highly fragmented and inefficient. Creators must juggle multiple platforms for trends, music, and scripting, leading to wasted time, reduced productivity, inconsistent branding, and creative burnout. At the same time, they face a “compliance minefield,” where misunderstanding platform rules can result in low reach, content removal, or account penalties. Although many aim to turn ideas into viral content, a growing creation bottleneck persists, further worsened by existing AI tools that lack brand memory and produce generic, inconsistent outputs—ultimately causing a strategic engagement gap where content fails to authentically connect with audiences.

- **Importance of Solving This Issue:** Content Buddy solves the "Pre-Production Bottleneck" through its strategic director capability powered by Artificial Intelligence, which clones the creator's brand DNA and transforms hours of messy research into a cohesive and platform-friendly production process. By speeding up the content creation process within a few seconds while maintaining its authenticity, the creator is able to clearly state the intended message and produce a highly shareable piece of content without having to seek help from a professional marketing agency.

- **Strategic Fit / Impact:** Content Buddy connects to the AI-Native Creator Ecosystem, leveraging the Z.ai GLM's ability to process multimodal data and translate unstructured creative ideas into a structured production asset. With the ability to generate "Platform-Aware" intelligence, Content Buddy can act as a key tool that helps entrepreneurs in navigating the intricacies of social media algorithms.

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3. Product Purpose

3.1. Main Goal of the System: The basic aim of Content Buddy is to empower everyone to create great quality content using an all-encompassing artificial intelligence strategist. With Content Buddy, users get to overcome the expensive hiring of external agencies and marketing specialists through its personal scriptwriting, visual storyboard creation that includes camera angles and transition edits. It will also make sure that it follows the guidelines set forth by TikTok, Instagram, and YouTube, so there are no issues with copyright laws.Content Buddy offers maximum convenience when it comes to producing professional quality videos at affordable prices and increased efficiency.

3.2. Intended Users (Target Audience):

- Business and Product Teams
- Founder of Products (Early Stage)
- UI/UX Designer
- Individual Content Creators
- Small-to-Medium Enterprises
- Solo Professionals
- E-commerce Sellers
- Development Team
- QA Team
- Students Organizations & Campus Clubs

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4. System Functionalities

4.1. Description: Content Buddy is a quick and powerful Creative Intelligence Engine that transforms the user's intentions to ready-to-produce social media content. Using Style Cloning to Script approach, it analyzes past video URLs to learn a creator's unique language, pacing, and engagement patterns, then uses these insights to generate coherent, personalized scripts and content strategies that maintain authenticity while maximizing audience engagement.

4.2. Key Functionalities:

- **Signature Style Learning:** Examines three external videos in order to understand the overall vibe and caption types used to ensure that new ideas feel like true originals.
- **Visual Storyboarding:** Creates scene visuals to prepare for filming the content and help define the overall concept.
- **Multi-Perspective Scripting:** Produces both an **actor version** (dialogue and delivery) and a **director version** (camera angles, lighting, and transitions) for complete production guidance.
- **Intelligent Content Calendar:** A constant calendar icon that shows in the user interface allows users to manage the global content pipeline and visualize all upcoming titles and dates.
- **Precision Posting Notifications:** Sends timely alerts to maximize reach and posting efficiency.
- **Multimodal Asset Generator:** Automates the generation of an entire production package, including suggested title, popular songs for use, and optimal captions and scripts.
- **Centralized Navigation & History (Modern UI/UX)**
  Clean interface with persistent side, top, and bottom panels, dark/light mode, and easy access to past content and settings.

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- **Compliance & Safety Engine (Copyright Reminder):** Integrated safeguard within Content Buddy, designed to analyze the selected platform and its specific policies to ensure that users remain compliant with platform policies and reduces the risk of copyright infringement or content removal.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: We chose to employ Z AI's GLM-5.1-flash model due to its speed and the ability to produce formatted JSON output for the reliability of our dynamic UI elements (i.e., Content Calendar, Compliance Check). With a multi-step prompting process and the ability to reason about user intent of up to 500 characters while semantically cloning URLs provided by the user, the model runs a dedicated Demo Mode fallback whenever API latencies occur.

4.3.2. Prompting Strategy: Multi-Step Agentic Prompting. In order to effectively turn users' raw ideas into professional productions, Content Buddy uses the Multistep Agentic Prompting strategy in its unique relay race of AI logic. Unlike traditional prompting systems that are limited to one prompt, Content Buddy splits the work into three distinct steps: Research Agent who clones the user's unique "Sonic DNA" from URLs, Creative Director Agent who produces high-quality scripts/storyboards in that exact style and tone, and Guardian Agent who checks whether there could be any copyright issues or platform-specific shadowbans.

4.3.3. Context & Input Handling: Our system will accept input for a topic up to 500 characters and three video links. If you feed our system more than that, for example, when copying an extensive note, we don't simply truncate your text halfway through the sentence. Rather, we leverage the AI's "super brain" to extract the key terms and objectives from your input. When there are too many terms to cover, we automatically summarize your inputs to create a concise instruction to guide the AI. Although we have a generous limit of 50 million tokens, we still enforce such limitations to ensure that the AI produces a comprehensive plan within eight seconds.

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4.3.4. Fallback & Failure Behavior: Content Buddy employs a Circuit Breaker and Graceful Degradation approach. The application will stop waiting for an answer from the AI if there isn't one for five seconds and will save the current progress of the user to ensure no data is lost. Instead of an unfriendly and unappealing Error 500 message, users see either a notification or are put into Demo Mode – a mode where pre-saved data helps maintain all storyboards and calendars in excellent condition. Thus, despite any possible issues with internet connection or AI availability, the platform never looks amateurish and continues working normally, ensuring the best experience. In case of total API failure or internet disconnection, the application will switch over to a high-quality and pre-tested offline database and make sure that script generation, storyboards, and content calendars continue functioning and looking great at the time of critical presentations.

5. User Stories & Use Cases

- **User Stories:** As an Individual Content Creator, I want to provide URL links of my previous work so that the AI can learn my style and generate new scripts and storyboards. As a Small-to-Medium Enterprise (SME) owner, I want to generate professional storyboards and shooting angles automatically so that I can produce high-quality ads without hiring a dedicated film crew.
- **Use Cases (Main Interactions):**

  Prompt to UI Flow: Users login account and select platform used, then insert maximum 3 URL links, then enter the prompt, Z AI GLM parses the intent and extracts the pacing and tone, and get to provide a complete production package including recommend title, trending audio, captions, hashtag, script and storyboard.

  Flow Generation: User check copyright status and platform restriction based on the selected platforms, click next to display grid calendar view for better posting time.

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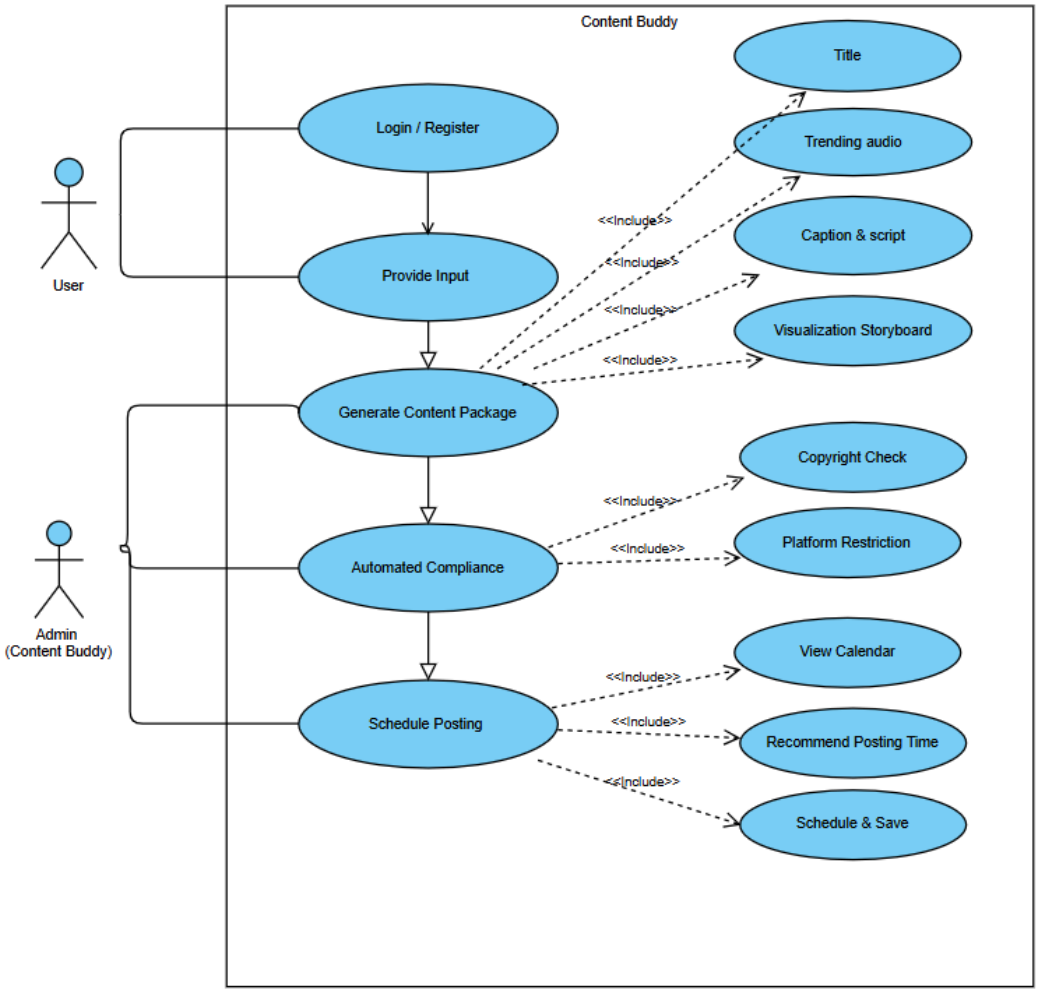
- **Use Case Diagram:**



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6. Features Included (Scope Definition)

- A workspace with layers that enable the user to see storyboard visuals, recommend title, trending songs and caption all in one non-linear space.
- A calendar feature which automatically schedules generated content into time slots of maximum engagement on Tiktok, Instagram, and Facebook.
- An embedded security tool that conducts a policy check on the platform to make sure that there is no violation that could lead to shadow bans and monetization strikes.
- Automated creation of an actor's version of the script and a director's version of the same script to cover camera movements and editing expertise.
- An automated extraction tool that can clone up to 3 links from the user and replicate the Hooks, Sonic DNA, and Pacing of their content.
- A posting notifications that sends timely alerts 10 minutes before the posting time to improve reach and consistency.
- A clean interface with side/top/bottom panels, dark/light mode, and easy access to history, settings, and user data.

7. Features Not Included (Scope Control)

- The platform generates scripts and storyboards but does not include a built-in video editor for assembling footage.
- Content cannot be automatically uploaded to TikTok, Instagram, or YouTube due to API and permission constraints; manual posting is required.
- Multiple users editing the same storyboard simultaneously (Live Collaboration) is currently excluded.
- While scripts and storyboards are generated, the system does not produce final AI-generated video files or 3D animations
- Subscription models and payment gateway integration

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8. Assumptions & Constraints

- **User Input:** The system cannot proceed without human input prompts. The user must introduce a maximum of 3 links of ideas provided and select the intended platform.
- **Internet Dependency:** Users must have an active internet connection for API calls and real-time content generation.
- **Accessible Input Data:** All URLs provided by users must be public and accessible for the system to analyze and extract style elements.
- **LLM Cost Constraint:** Although a top-grade tokenization of 50M Z.ai tokens is in place, it is designed to operate at scale. But we enforce a limit of 500 characters on the initial prompts to keep the AI aligned with the user's main objective and keep the generation time below 8 seconds.
- **Technical Constraints:** In order to create a quality visual guide, we employ the Pollinations API to produce a storyboard outline. An internet connection is necessary to synchronize the AI-written script with its visual output.
- **Performance Constraints:** For the best possible experience and user retention rate, the AI generation process is intentionally limited to a 4-scene storyboard. This limitation allows the AI to conduct a multi-agent reasoning process along with the visual sketching procedure within a latency of less than 8 seconds. It prevents “user drop-off” but provides enough material for a standard 30-60 second short form video production.

9. Risks & Questions Throughout Development

- **Homogenization of Creativity:** How can we prevent the AI from giving everyone the same recommendations and suggestions?
- **Drift in Policy:** Algorithms evolve on the platform every week. How frequently does the “Buddy Agent” algorithm need updating to stay current?
- **Relevance of Image:** What would happen if the Pollinations AI cannot put the highly specific shooting style in context within the sketch?

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